

DataCORE

Exploring Data with Code

Afternoon: DataCORE

- 1. Orientation — Tools & Environment**
- 2. Exploring Your First Dataset — NBA Home Advantage**
- 3. AI as a Coding Assistant**
- 4. Scientific Storytelling — Data Captions**

DataCORE isn't about software — it's about asking questions and finding answers in data.

1:00 – 1:20 · Orientation

Tools & Environment

DataCORE Orientation

- Everything runs in **Google Colab**, in your browser — **no installation**.
- You'll **open a notebook**, **run a cell**, and see output instantly.
- Today's first cell just says hello →

Big idea: DataCORE is *not* about software. It's about **asking questions and finding answers in data**.

Python

R

```
print("Hello, Research!")
```

Hello, Research!

1:20 – 2:20 · Exploring Your First Dataset

NBA Home Advantage

Step 1 · Load & peek at the data

Open the shared notebook, then **look before you leap** — preview the rows and the shape.

Python

R

```
import pandas as pd
df = pd.read_csv("nba_home_away.csv")
print(df.head())          # first 5 rows
```

	date	home_team	away_team	home_pts	away_pts	home_wins	away_wins
0	2023-10-24	Atlanta Hawks	Minnesota Timberwolves	83	106	0	1
1	2023-10-24	Atlanta Hawks	Cleveland Cavaliers	147	126	1	0
2	2023-10-24	Atlanta Hawks	Boston Celtics	101	107	0	1
3	2023-10-24	Atlanta Hawks	Sacramento Kings	114	120	0	1
4	2023-10-24	Atlanta Hawks	Houston Rockets	102	99	1	0

```
df.shape          # rows = games, cols = fields
```

(1230, 7)

❓ How many games are in the data? How many teams?

Step 2 · Basic statistics at a glance

Python R

```
print(df.describe().round(2))    # summary stats per column
```

	home_pts	away_pts	home_wins	away_wins
count	1230.00	1230.00	1230.00	1230.00
mean	111.54	110.30	0.58	0.42
std	14.72	11.07	0.49	0.49
min	60.00	75.00	0.00	0.00
25%	102.00	103.00	0.00	0.00
50%	112.00	110.00	1.00	0.00
75%	122.00	118.00	1.00	1.00
max	163.00	149.00	1.00	1.00

 One line gives you the **shape of the whole dataset** — means, spread, and extremes.

📊 Step 3 · Calculate the home win rate

`home_wins` is **1** when the home team won and **0** when it lost — so its **average is the win rate**.

Python R

```
df["home_wins"].mean()    # share of games won by the home team
```

```
np.float64(0.5829268292682926)
```

🌀 Is it above 0.50? By how much? That's the **home advantage** in one number.

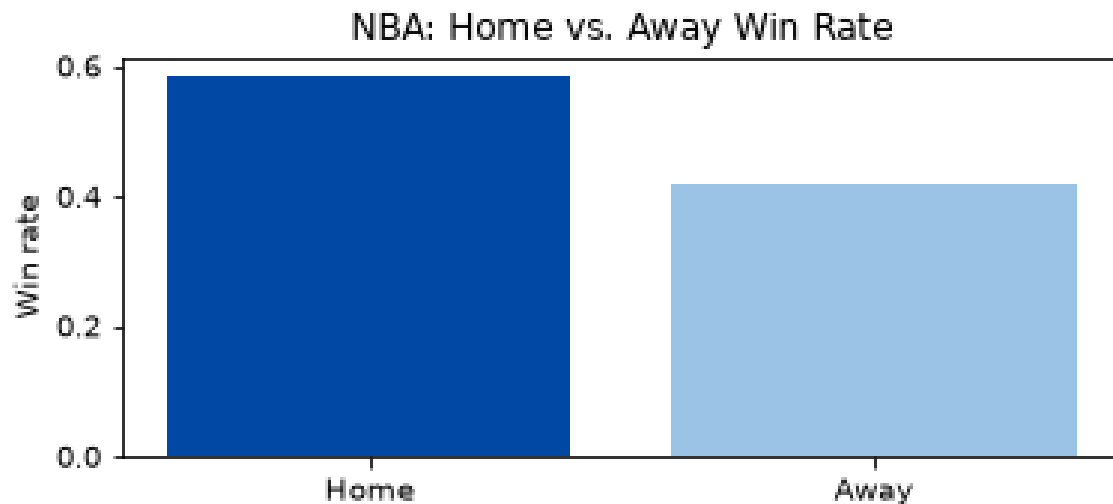
Step 4 · Make a bar chart

Python R

```
import matplotlib.pyplot as plt

home = df["home_wins"].mean()
away = 1 - home

plt.figure(figsize=(6, 2.4))
plt.bar(["Home", "Away"], [home, away],
        color=["#0148A4", "#9bc3e6"]) # color the bars
plt.ylabel("Win rate")
plt.title("NBA: Home vs. Away Win Rate") # title & labels
plt.show()
```



2:20 – 2:45 · AI as a Coding Assistant

You asked the AI for a "win percentage" column...

It instantly wrote this. You run it — and the numbers look **way too high** (over 100%!). Can you spot the bug?

Python R

```
# AI: "what % of games did the home team win?"  
home_win_pct = df["home_wins"].sum() / df["away_wins"].sum() * 100    # ÷ away wins  
home_win_pct
```

```
np.float64(139.76608187134502)
```

 That gives **139.8%** — home wins should be divided by **total games**, not away wins.

✔ Fix it: divide by total games

Python R

```
home_win_pct = df["home_wins"].sum() / len(df) * 100  
home_win_pct
```

```
np.float64(58.292682926829265)
```

Key lesson: AI *writes* code — **humans verify it**. You have to understand what the code does.

2:45 – 3:00 · Scientific Storytelling

Data Captions

Write a 2-sentence caption for your chart

1. What does the chart **show**?
2. What is **surprising** or notable?

Use **precise language** — exact numbers, clear comparison.

 Read your caption to your **partner**. Is every number specific and correct?

*"Home teams won **58.3%** of games in our dataset, compared to **41.7%** for away teams. The edge is real, but smaller than the roar of a home crowd on TV might suggest."*